

Assignment - 2

- ★ Web 1.0 (1990s - early 2000s) : Read only web
- static website with limited interactivity
 - user can mainly read information but can't contribute
 - Focused on sharing information rather than creating.

- ★ Web 2.0 (mid-2000s - present) The social web
- interactive website that encourage user participation.
 - Emphasis on user generated content, social networking & collaboration.
 - Rise of social media platforms, blogs, video sharing sites
 - e.g. Facebook, twitter, youtube.

- ★ Web 3.0 (emerging) :- The semantic web :-
- decentralized web with focus on data semantic & machine learning.
 - intelligent website that can understand & interpret user intent.
 - increased use of AI, blockchain & IOT.
 - e.g. decentralized finance, NFTs, metaverse platforms.

- ★ Request response for bvmengineering.ac.in :-

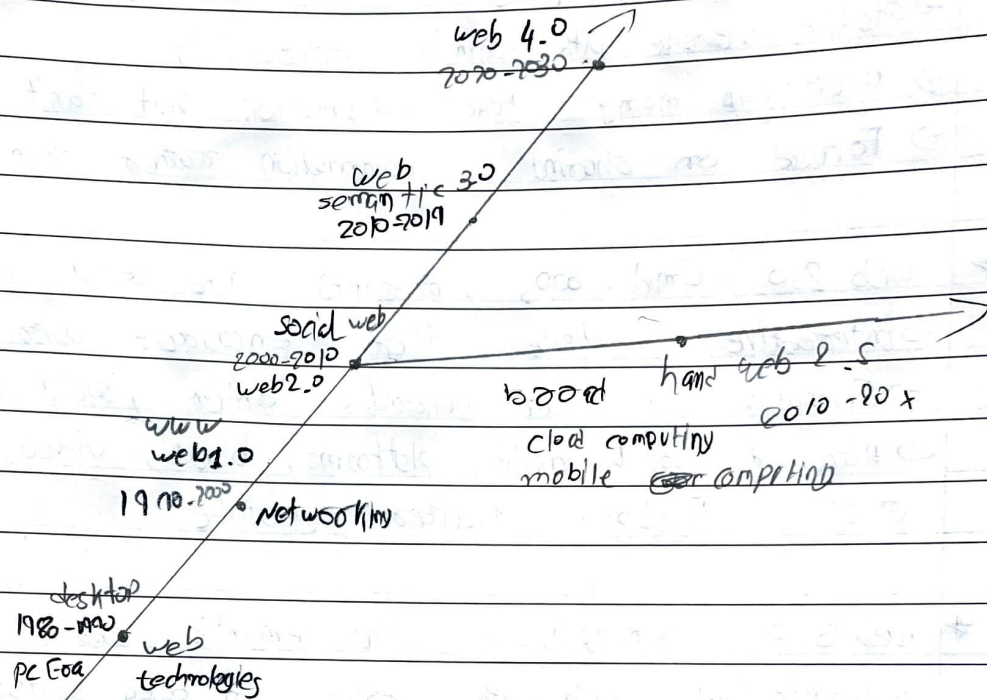
→ \$nslookup bvmengineering.ac.in

server unknown

Address :- 202.124.241.241

- ★ W3C : world wide web consortium

→ it is international community where member organizations work together to develop web standards.



→ Key Goals :-

→ To -lead world wide web to its full potential.

→ create web standards that ensures long term growth of web.

→ Key Activities :-

- developing web standards :- include specification for HTML, CSS, XML
- Promoting adoption of web standards.
- through education outreach & testing.
- Building tools & resources.

→ Impact :-

- W3C played crucial role in web development & growth of web. Their standards ensures that web is accessible to everyone & that it continues to evolve in a sustainable & interoperable way.

★ ARPANET :-

→ First operational packet switching network.

→ Advanced research projects Agency networks.

→ United states defense Advanced Research project Agency (DARPA) developed in the late 1960s

→ Goal is to enable sharing of research data among university & research institutions

→ Technology :- utilized Packet switching, where data is broken down to smaller packets that can be routed independently through network.

* Significance :-

- foundation for internet.
- ARPANET ~~is~~ did groundwork for development of TCP/IP.
- Fundamental suite of protocols that govern internet communication.

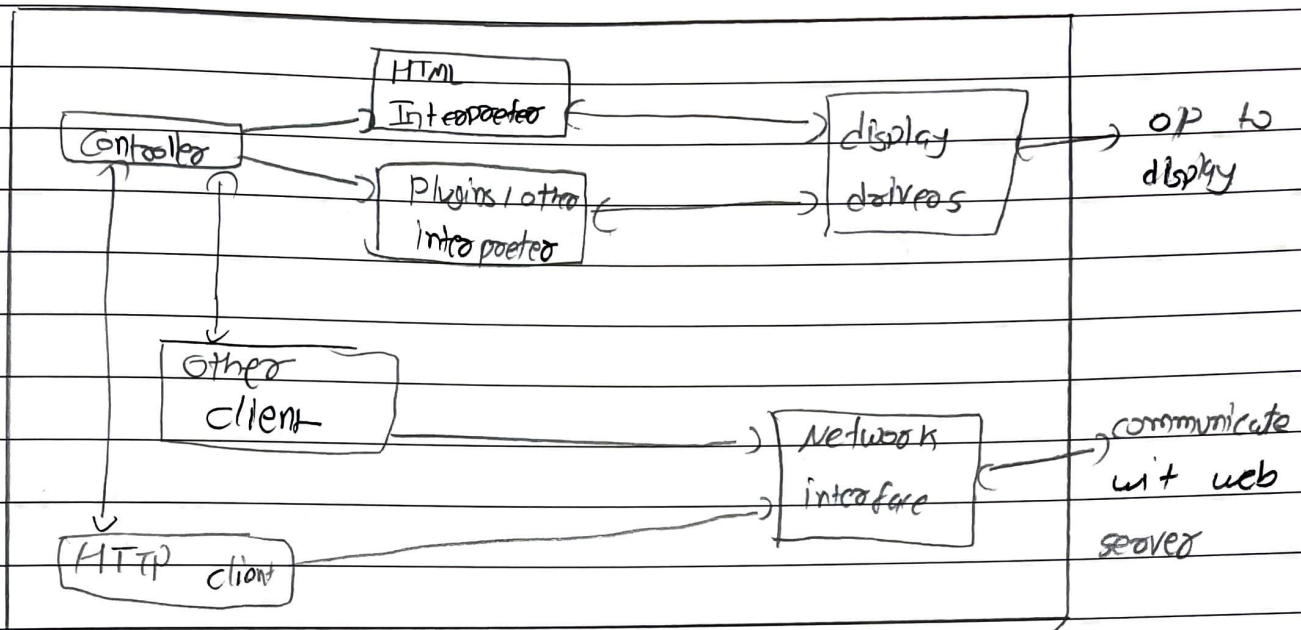
* Pioneer Research :-

→ served as testbed for early networking technologies & paved the way for future developments in computer networking.

* Popular tools :-

- File transfer Protocol :-
Filezilla, Cyberduck, WinSCP.
- Hyper text transfer protocol :-
web browsers, wget, curl.
- Secure shell :-
PuTTY, OpenSSH, Terminal.

* Web browser architecture.



→ **Controller**: central part of web browser, responsible for managing entire process, receive i/p from user.

→ **HTML interpreter**: responsible for parsing HTML code received from server & converts into structural representation that browser can understand.

→ **display devices**:
handle o/p to display screen, take rendered content from HTML interpreter and display it on screen.

→ **Network interface**:
- responsible for communication with web server.
- it sends HTTP request to server & receive HTML content in response.

→ **Plug-ins / other interpreters**:
used to handle specific type of content.

→ **HTTP client**:
if make HTTP request to web server.